

EC2091E – ANALOG CIRCUITS LAB

Department of Electronics & Communication Engineering

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ANALOG CIRCUITS MINI PROJECT
BUCK CONVERTER

Submitted by:

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Aim

The aim of this mini project is to design and implement a Buck Converter that steps down a DC input voltage of 9V to a regulated DC output voltage of 5V. The converter should maintain stable output voltage under varying load conditions while ensuring efficient power conversion.

Synopsis

A Buck Converter is a type of DC-DC switching converter used to step down a higher DC input voltage to a lower regulated DC output. Unlike linear regulators, which reduce voltage by dissipating excess energy as heat, a buck converter uses high-frequency switching along with energy storage components to achieve high efficiency.

In this project, the aim is to convert 9V DC to a stable 5V DC output using a manually controlled analog feedback system instead of a dedicated switching regulator IC.

Working Principle of Buck Converter

The buck converter operates by rapidly switching a P-Channel MOSFET (IRF9540N) between ON and OFF states.

- **When the MOSFET is ON:** Current flows from the supply through the MOSFET and into the inductor, storing energy in its magnetic field.
- **When the MOSFET is OFF:** The inductor releases the stored energy to the load through the Schottky diode (1N5819), ensuring continuous current flow.

The average output voltage is determined by the duty cycle D :

$$V_{out} = D \cdot V_{in}$$

$$V_{in} = 9V, \quad V_{out} = 5V, \quad D = \frac{\text{ON time}}{\text{Total switching time}}$$

Analog Feedback Regulation

To ensure that the output remains stable at 5V under varying load conditions, a closed-loop feedback control system is implemented.

Voltage Reference: TL431A

TL431A provides a precise 2.495V reference, which acts as the comparison baseline.

Error Amplifier: LM358

The output voltage is sensed using a resistor divider and compared with the TL431 reference using an LM358 operational amplifier.

- If the output voltage drops below 5V, the error voltage increases, increasing MOSFET duty cycle and raising output voltage.
- If the output voltage rises above 5V, the error voltage decreases, reducing duty cycle and bringing the output back to 5V.

This provides automatic closed-loop regulation.

Switching Device: IRF9540N (P-MOSFET)

Since a P-Channel MOSFET is used, gate-to-source voltage controls conduction:

MOSFET State	Gate Voltage Relative to Source	Result
Gate pulled LOW	Lower than Source	MOSFET ON
Gate pulled HIGH	Near Source voltage	MOSFET OFF

The LM358 output controls the gate and therefore the duty cycle.

Energy Storage Components

Component	Value	Function
Inductor	100 μ H	Stores energy during MOSFET ON time and ensures smooth current flow
Diode (1N5819)	Schottky	Provides current path during OFF time with low forward voltage drop
Output Capacitor	100 μ F	Reduces ripple and stabilizes output voltage

Why Manual Feedback Instead of IC?

Using discrete analog control enables:

- Understanding how feedback maintains voltage regulation
- Visualizing duty cycle variation under different loads
- Learning how op-amp based error amplifiers operate
- Observing real-world switching converter behavior instead of using integrated regulator ICs

This project demonstrates the core working principle of efficient DC power conversion through regulated switching and controlled energy transfer.

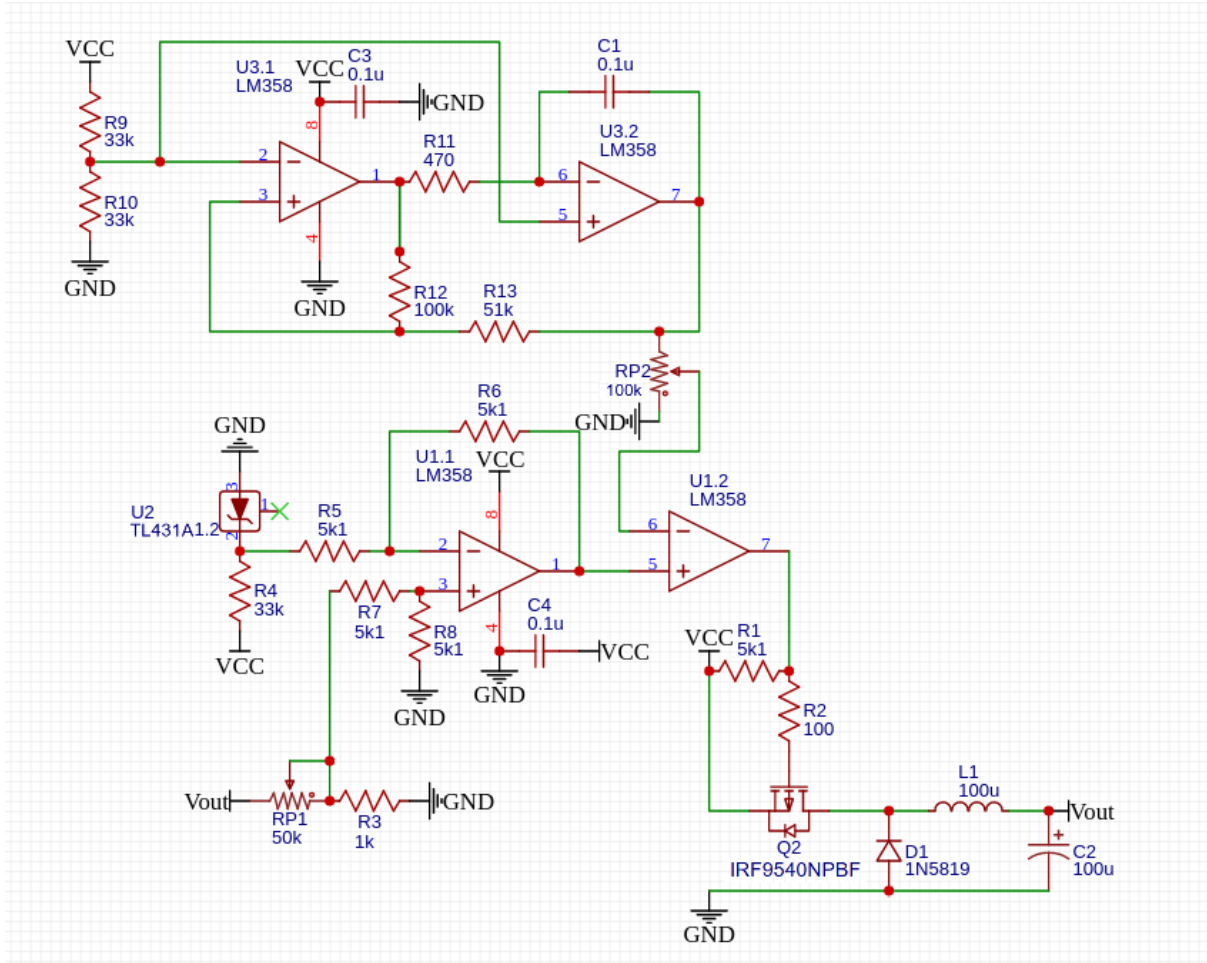


Figure 1: Buck Converter Reference Design Circuit Diagram

Segments in the Buck Converter Circuit

The complete buck converter circuit can be divided into three major functional segments: the relaxation oscillator, the feedback differential stage, and the power stage. Each part performs a specific role in generating, regulating, and delivering the required 5V output from a 9V input supply.

1) Relaxation Oscillator

The buck converter requires a periodic switching signal to turn the MOSFET ON and OFF. In this circuit, the switching signal is generated using a relaxation oscillator built using the LM358 operational amplifier. The oscillator produces a square wave at a fixed switching frequency, which determines how rapidly the MOSFET transitions between conduction and cutoff.

The principle of operation is based on the repeated charging and discharging of a capacitor through a resistor network. When the capacitor voltage reaches a threshold, the op-amp output switches polarity, reversing the charging direction. This continuous cycle results in a stable periodic waveform.

The switching frequency is chosen such that:

- The inductor has adequate time to store and release energy during each cycle.
- Output ripple remains low.
- Switching losses are minimized.

Typically, this frequency lies in the tens to hundreds of kilohertz range, depending on the MOSFET and inductor characteristics.

Key Point: The relaxation oscillator only generates the switching signal. It does not regulate the output voltage. Regulation is achieved by the next stage.

2) Feedback Differential (Error Amplifier) Stage

While the oscillator controls the switching frequency, the feedback stage controls the duty cycle, directly determining the output voltage.

The output voltage is sensed using a resistor divider and compared with a precise reference voltage generated by the TL431A, which functions as a stable 2.5V adjustable reference. This ensures far better accuracy and temperature stability than a normal zener diode.

The sensed output voltage and the TL431 reference are fed into a differential amplifier built using LM358.

- If the output voltage drops below 5V: the error voltage increases, the duty cycle increases, the MOSFET remains ON for a longer duration, and the output voltage rises.
- If the output voltage rises above 5V: the error voltage decreases, the duty cycle reduces, the MOSFET turns OFF sooner, and the output voltage decreases.

This creates a closed-loop automatic regulation mechanism, allowing the converter to maintain a stable 5V output even under varying load conditions.

This stage acts as the control system of the converter.

3) Power Stage

The power stage is responsible for actual energy transfer to the load. It consists of the switching device, the inductor, diode, and output capacitor.

Part	Component Used	Purpose
Switching Element	IRF9540N P-Channel MOSFET	Controls when energy is delivered to the inductor
Energy Storage	100 μ H Inductor	Stores and releases energy while smoothing current
Freewheeling Path	1N5819 Schottky Diode	Provides current path when the MOSFET is OFF
Output Filtering	100 μ F Capacitor	Reduces voltage ripple and stabilizes output

Operation

- **When MOSFET is ON:** Current flows from the supply through the MOSFET and inductor to the load, and energy is stored in the inductor.
- **When MOSFET is OFF:** The inductor releases stored energy, the diode conducts to keep current continuous, and the capacitor filters ripple to maintain steady 5V output.

Thus, the power stage performs the actual voltage conversion and energy delivery, while the relaxation oscillator and feedback stages determine how and when this power transfer occurs.

Component	Part Number / Value	Role / Purpose in Circuit
P-Channel Power MOSFET	IRF9540N	Used as the high-side switching device. Its P-channel configuration allows simpler gate-drive control for switching the input supply to the inductor.
Operational Amplifier	LM358 (Dual Op-Amp)	One op-amp section is used to build the relaxation oscillator for generating the switching signal, and the second section is used as the error/differential amplifier for output voltage regulation.
Programmable Precision Reference	TL431A	Provides a stable 2.5V bandgap reference. Functions similar to a high-accuracy zener diode to compare the sampled output voltage in the regulation loop.
Schottky Diode	1N5819	Acts as the freewheeling diode, allowing current to continue flowing when the MOSFET is OFF, while minimizing switching and forward conduction losses.
Inductor	100 μ H	Stores energy when the MOSFET is ON and releases it when OFF, ensuring smooth and continuous current to the load.
Output Capacitor	100 μ F (Electrolytic)	Filters and reduces output ripple to maintain a stable 5V DC output.
Decoupling / Bypass Capacitors	0.1 μ F, 10 μ F	Used for noise suppression and improving power supply stability around active components like op-amps and the reference IC.
Resistors	Various Values	Used in the feedback divider network, to set reference scaling, and for stabilization and biasing purposes.
Potentiometers	100 k Ω (Adjustable)	Allows fine tuning and precise adjustment of the output voltage to exactly 5V.

Table 1: List of Components Used in the Buck Converter Circuit

Design Calculations (Reduced)

For a Buck Converter:

$$V_{out} = D \cdot V_{in}$$

Given:

$$V_{in} = 9\text{V}, \quad V_{out} = 5\text{V}$$

$$D = \frac{V_{out}}{V_{in}} = \frac{5}{9} \approx 0.56$$

Inductor Ripple Current

$$\Delta I_L = \frac{(V_{in} - V_{out}) \cdot D}{L \cdot f_s}$$

Using:

$$L = 100 \mu\text{H}, \quad f_s \approx 100 \text{ kHz}$$

$$\Delta I_L = \frac{(9 - 5) \cdot 0.56}{100 \mu\text{H} \cdot 100 \text{ kHz}} \approx 0.22 \text{ A}_{pp}$$

CCM Boundary

$$I_{\min, \text{CCM}} = \frac{\Delta I_L}{2} \approx 0.11 \text{ A}$$

So the converter remains in continuous conduction when load current is above $\approx 110 \text{ mA}$.

Output Ripple (Approximate)

$$\Delta V_{out} \approx \Delta I_L \cdot R_{ESR}$$

For a low-ESR output capacitor:

$$\Delta V_{out} \approx 10 - 20 \text{ mV}$$

Quick Takeaway

- Duty cycle ≈ 0.56
- Inductor ripple $\approx 0.22 \text{ A}$
- Stays in CCM above 110 mA
- Use low-ESR capacitor to maintain $\approx 10\text{--}20 \text{ mV}$ ripple

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Working

Working (Step-by-Step, in Three Segments)

Segment 1 — Relaxation Oscillator (LM358)

An LM358 section is configured as a Schmitt-trigger relaxation oscillator to generate a periodic ramp/triangle waveform that sets the time base for PWM. A timing capacitor charges and discharges through a resistor network while positive feedback on the op-amp creates two thresholds (upper and lower). When the capacitor voltage crosses the upper threshold, the op-amp output toggles low; when it crosses the lower threshold, the output toggles high. This produces a clean, near-linear ramp at the capacitor and a corresponding square wave at the op-amp output. The oscillator frequency is set by the RC network and the hysteresis ratio. This carrier waveform is intentionally independent of load or input voltage so the duty-cycle alone controls the output power.

Segment 2 — Feedback & Differential Amplifier (LM358 + TL431A 2.5 V “Zener” Reference)

Output voltage is sensed through a resistor divider and compared against a stable 2.5 V reference produced by a TL431A. In this design, the TL431A is used as a precision “zener” regulator of 2.5 V, giving a low-drift band-gap reference instead of a simple diode zener. The second LM358 section acts as a differential/error amplifier: it subtracts the scaled V_{out} from the 2.5 V reference and provides an amplified error signal. A small RC network around the amplifier (compensation) shapes its response so the loop remains stable across input and load changes. A front-panel potentiometer is placed either in the divider or in the error-amp gain path to allow manual trimming of the regulated output near 5 V. In steady state, the error goes to (near) zero, meaning the divided V_{out} equals 2.5 V and the loop automatically finds the duty-cycle that holds $V_{out} = 5$ V.

Segment 3 — Comparator / PWM Modulator → High-Side Power Stage (P-MOSFET IRF9540N) → LC Output

The oscillator’s ramp (carrier) and the error-amp output (control) feed a comparator to generate a pulse-width-modulated (PWM) gate signal: when the ramp is below the control level the comparator output is active, and when above it the output is inactive. That PWM drives a P-channel MOSFET (IRF9540N) used as a high-side switch from the 9 V input to the inductor. Gate-drive resistors and a pull-up/pull-down network ensure fast, clean transitions and protect against shoot-through or false turn-on.

When the P-MOSFET is ON, energy flows from the 9 V source into the inductor and the output capacitor, raising the inductor current. When it turns OFF, the inductor current freewheels through the catch (Schottky) diode into the load and output capacitor, maintaining current continuity. The L-C filter (inductor plus low-ESR capacitor) smooths the chopped current into a low-ripple 5 V DC output. Resistors elsewhere serve current limiting, sensing, and bias duties, while decoupling capacitors keep the op-amp and reference rails quiet. The feedback path from V_{out} back to the differential amplifier closes the loop: any disturbance that nudges V_{out} up or down is sensed and immediately corrected by an opposite change in PWM duty-cycle.

Challenges and Design Considerations

During the development of this buck converter, we faced several practical challenges, especially in separating the control circuitry from the power stage. Initially, we powered the control section directly from the 9 V input supply. This caused the reference levels and the feedback amplifier to shift whenever the input or load changed, leading to unstable duty cycle control. To solve this, we decided to power the control circuitry from the regulated output itself. Once the buck began regulating, the control stage also received a stable supply, which made the feedback loop much more consistent.

We also had to carefully manage signal levels between stages. The op-amp outputs and the MOSFET gate need clean logic transitions, so additional pull-up and pull-down resistors were added to ensure reliable switching and noise immunity. These small passive corrections made a noticeable difference in reducing unwanted gate floating and jitter in the PWM waveform.

A major learning experience was working with analog feedback instead of relying on regulator ICs or microcontroller-based PWM control. With analog control, the duty cycle must respond naturally to the error voltage, and we had to understand how the LM358 and TL431 interact to maintain stability. This required experimenting with the oscillator behavior, reference levels, and feedback gain before the output settled properly. Although harder, this approach gave us a deeper understanding of how op-amps behave inside real closed-loop power systems.

In terms of limitations, the circuit performs well at 9 V input, but as the input voltage increases, the LM358 cannot drive the gate of the IRF9540N strongly enough. The op-amp's output swing is limited and cannot pull the MOSFET gate low enough relative to its source at higher voltages to fully turn it ON. This is a known limitation when using a standard op-amp to directly drive a P-channel high-side MOSFET without a dedicated gate driver.

However, within its intended operating range, the converter is stable and reliable. It is well-suited for basic applications like powering microcontrollers, 5 V logic circuits, digital ICs, sensors, and embedded boards, especially when working from a battery or portable DC source. This makes the project a good demonstration and first-hand implementation of switching regulation, while also giving us a stable 5 V supply for use in other circuits outside the laboratory power environment.

Interesting Observation During Testing

While adjusting the potentiometer in the feedback network, we noticed that the circuit could be made to hold the output at a fixed level independent of switching, effectively turning the feedback stage into a unity-gain buffer / voltage follower. This was a useful side discovery because it demonstrated how the op-amp naturally tries to match its inputs under closed-loop conditions. This reinforced our conceptual understanding of op-amp feedback before applying it to PWM control.

Overall Outcome

Within its intended range, the converter provides a stable 5 V output and is well suited for powering microcontrollers, digital ICs, and small embedded circuits directly from a battery source. More importantly, the project served as a hands-on introduction to switching regulators, showing how component interactions, reference stability, and feedback control directly influence real converter behavior.

Results

The buck converter successfully stepped the 9V input supply down to a stable 5V output. The output remained steady under normal load conditions with only small ripple due to the inductor-capacitor filtering stage. The converter operated in continuous conduction mode for load currents above approximately 110 mA. After adjusting the feedback potentiometer, the output voltage matched the expected value. Additionally, the same feedback control allows the output to be reduced to lower voltages such as 3V when required.

Conclusion

This project demonstrated the working of a buck (step-down) converter using analog feedback control. The TL431 reference, LM358 error amplifier, and IRF9540N P-channel MOSFET together provided closed-loop duty-cycle regulation. The circuit performed reliably at a 9V input, providing a stable 5V output suitable for powering microcontrollers and digital circuits. While the design has limitations at higher input voltages due to op-amp gate drive constraints, it provided valuable insight into feedback stability, gate control challenges, and analog regulation behavior.

Overall, this mini project served as a solid introduction to switching power supplies, bridging theory with practical implementation and hands-on circuit debugging.